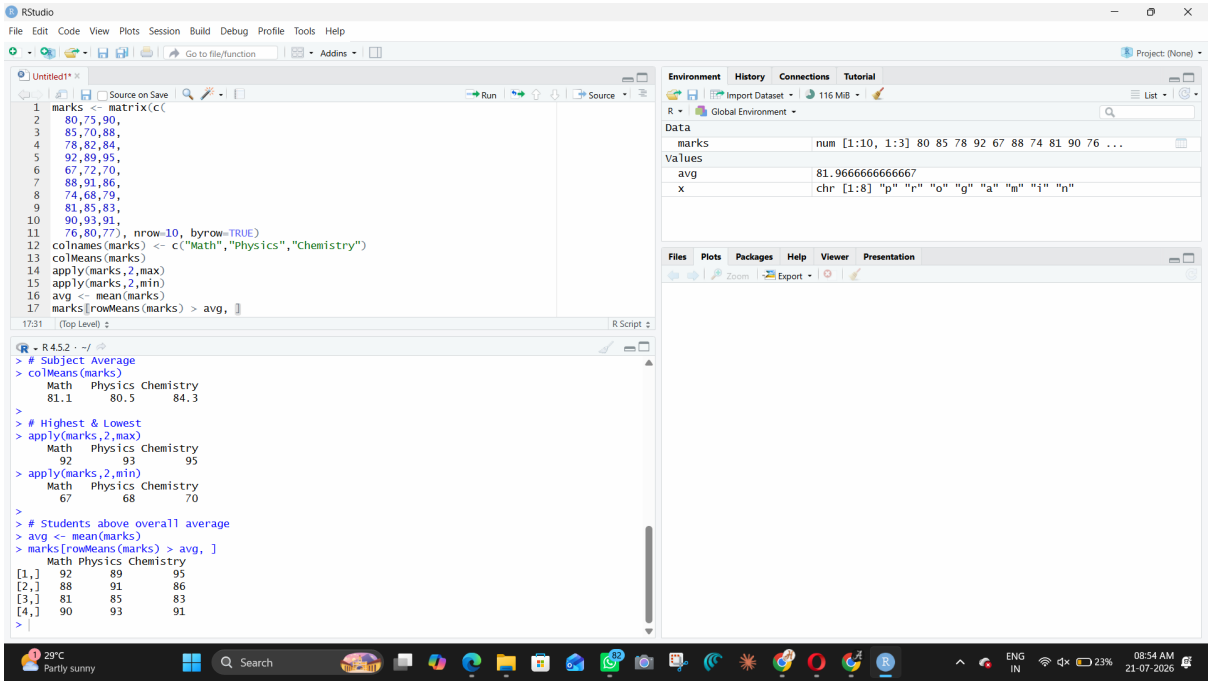
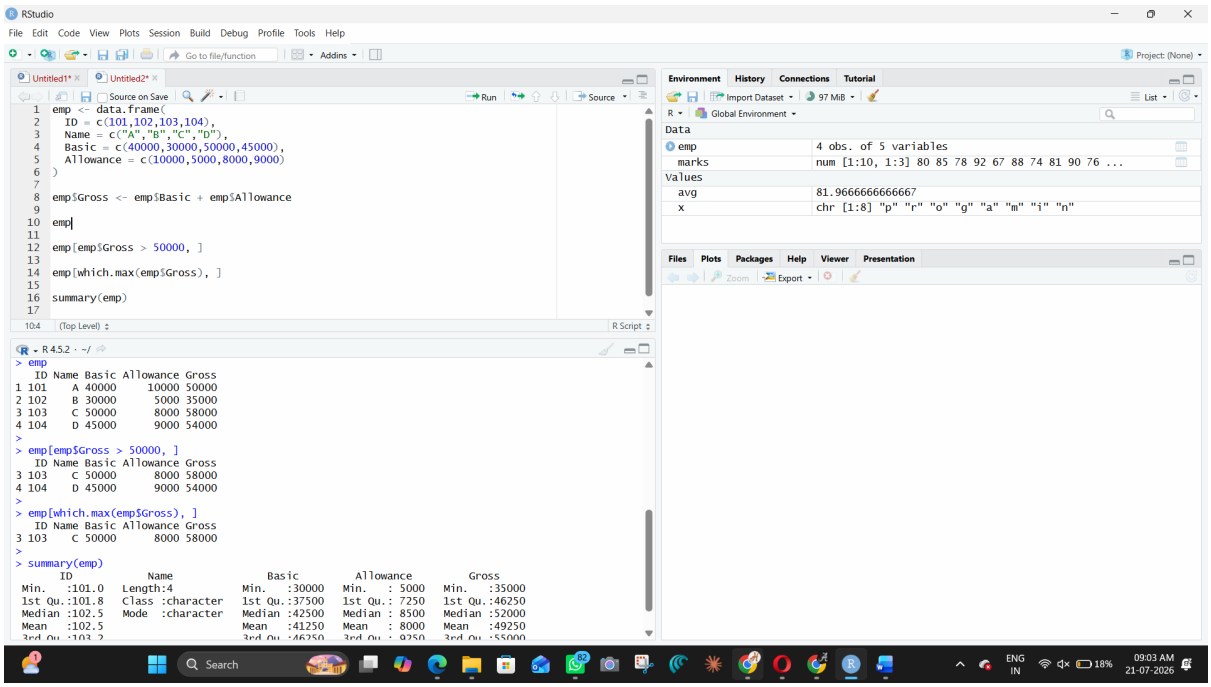
ITA0406 – R PROGRAMMING FOR STATISTICS

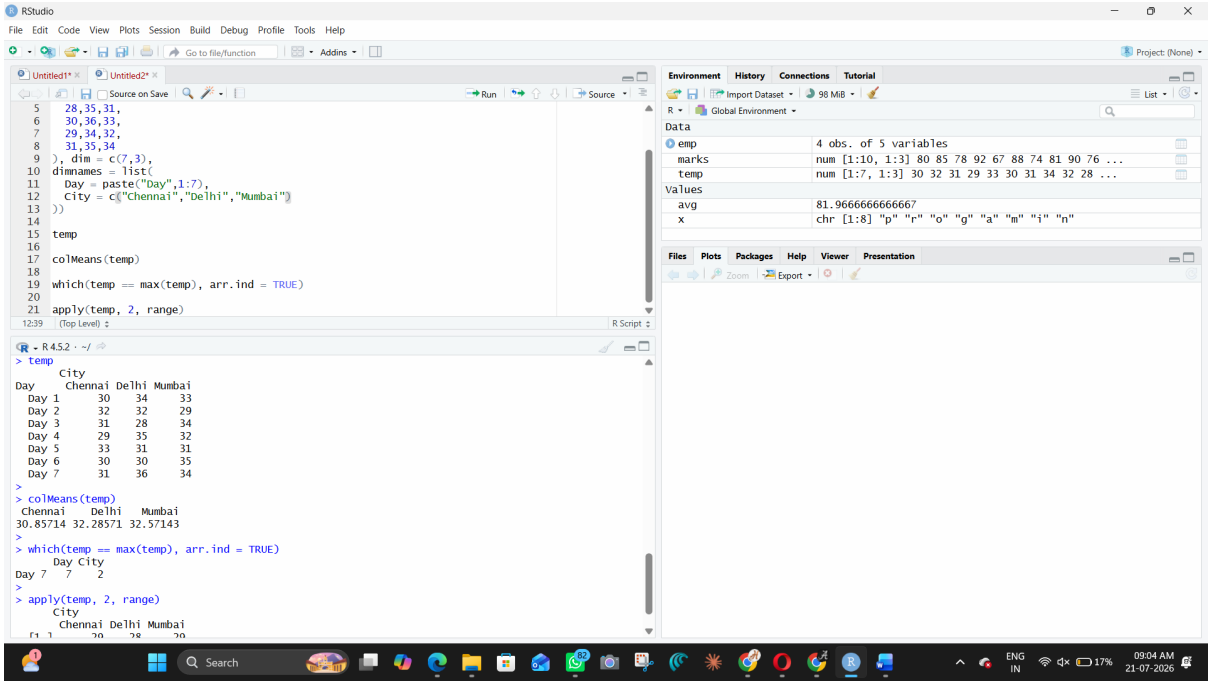
1)



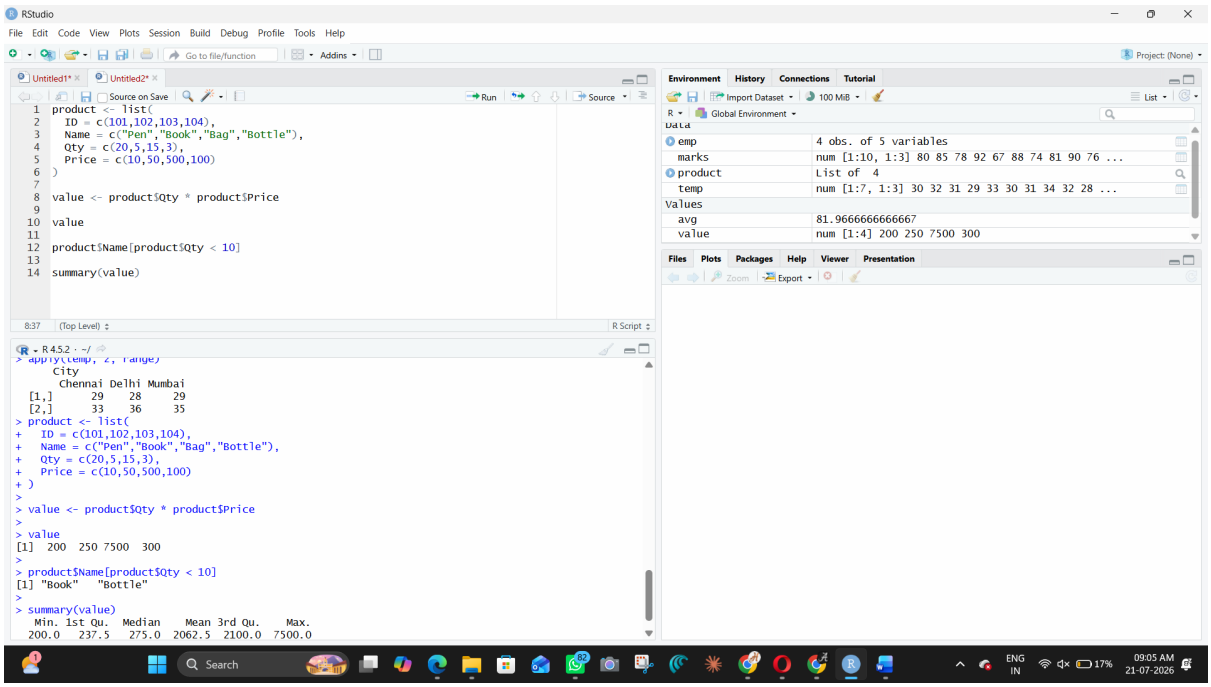
2)



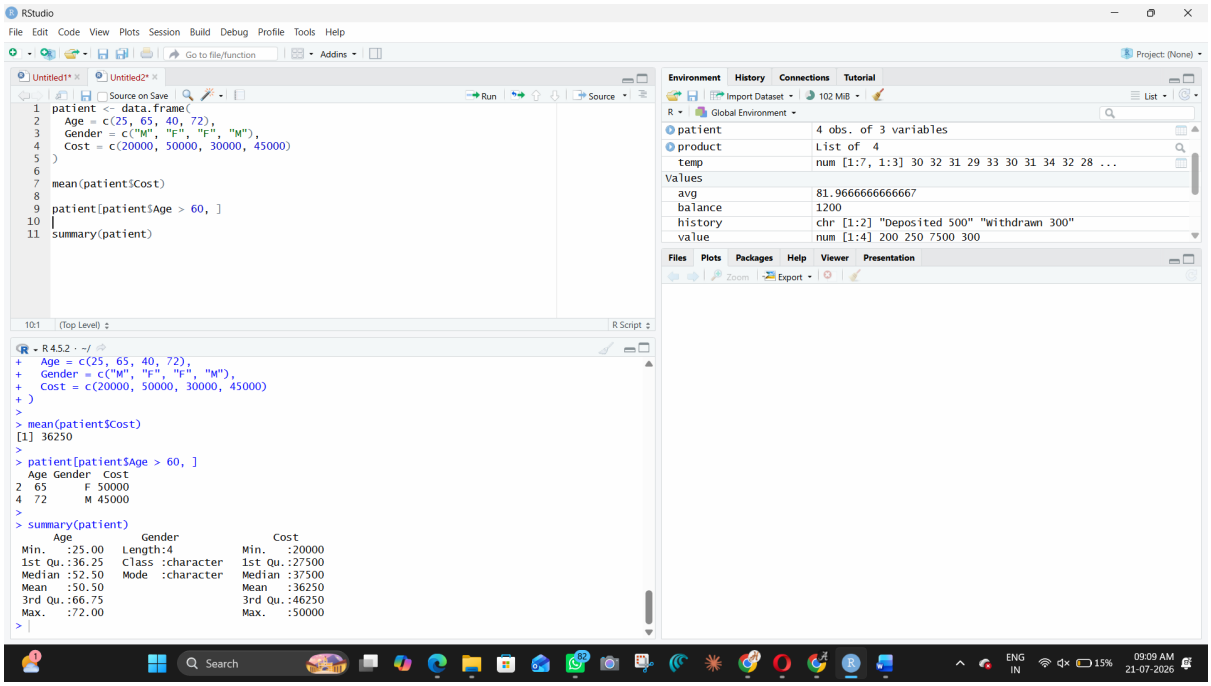
3)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
28,35,31,
30,36,33,
29,34,32,
31,35,34
), dim = c(7,3),
dimnames = list(
Day = paste("Day",1:7),
City = c("Chennai","Delhi","Mumbai")
))
temp
colMeans(temp)
which(temp == max(temp), arr.ind = TRUE)
apply(temp, 2, range)
Environment History Connections Tutorial
Data
emp 4 obs. of 5 variables
marks num [1:10, 1:3] 80 85 78 92 67 88 74 81 90 76 ...
temp num [1:7, 1:3] 30 32 31 29 33 30 31 34 32 28 ...
Values
avg 81.9666666666667
x chr [1:8] "p" "r" "o" "g" "a" "m" "i" "n"
Files Plots Packages Help Viewer Presentation
R 4.5.2 · ~/
> temp
City
Day Chennai Delhi Mumbai
Day 1 30 34 33
Day 2 32 32 29
Day 3 31 28 34
Day 4 29 35 32
Day 5 33 31 31
Day 6 30 30 35
Day 7 31 36 34
> colMeans(temp)
Chennai Delhi Mumbai
30.85714 32.28571 32.57143
> which(temp == max(temp), arr.ind = TRUE)
Day City
Day 7 7 2
> apply(temp, 2, range)
City
Chennai Delhi Mumbai

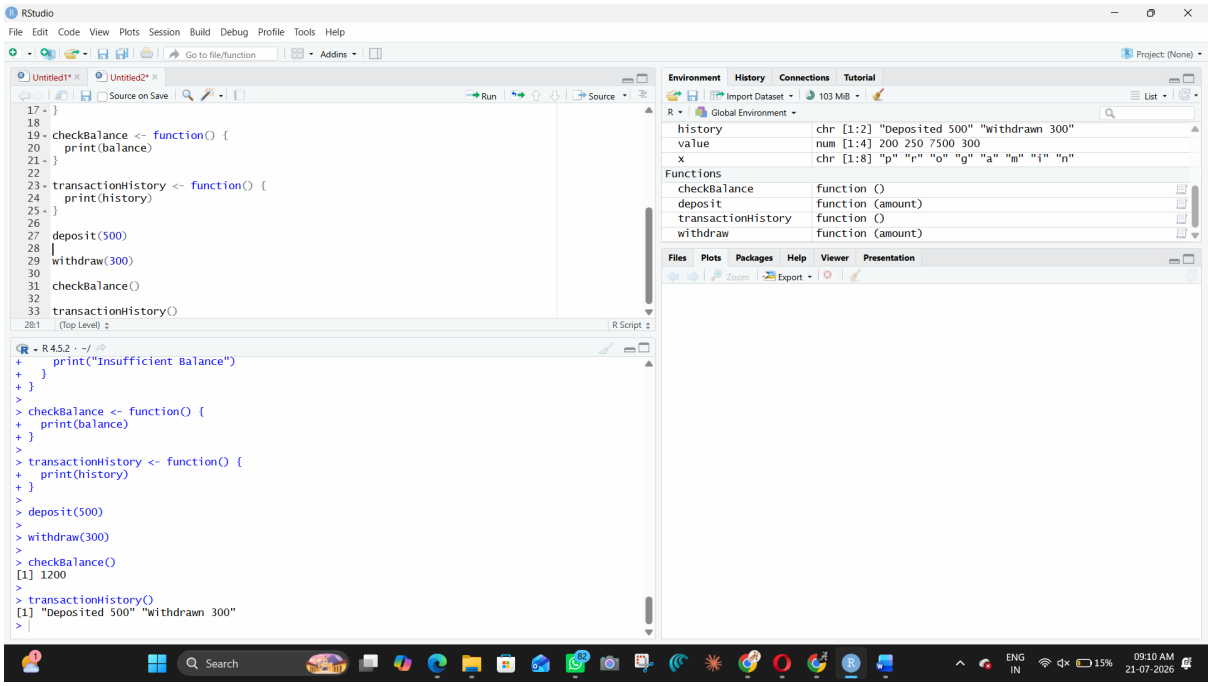
4)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
product <- list(
ID = c(101,102,103,104),
Name = c("Pen","Book","Bag","Bottle"),
Qty = c(20,5,15,3),
Price = c(10,50,500,100)
)
value <- product$Qty * product$Price
value
product$Name[product$Qty < 10]
summary(value)
Environment History Connections Tutorial
emp 4 obs. of 5 variables
marks num [1:10, 1:3] 80 85 78 92 67 88 74 81 90 76 ...
product List of 4
temp num [1:7, 1:3] 30 32 31 29 33 30 31 34 32 28 ...
Values
avg 81.9666666666667
value num [1:4] 200 250 7500 300
Files Plots Packages Help Viewer Presentation
R 4.5.2 · ~/
City
Chennai Delhi Mumbai
[1,] 29 28 29
[2,] 33 36 35
> product <- list(
+ ID = c(101,102,103,104),
+ Name = c("Pen","Book","Bag","Bottle"),
+ Qty = c(20,5,15,3),
+ Price = c(10,50,500,100)
+)
> value <- product$Qty * product$Price
> value
[1] 200 250 7500 300
> product$Name[product$Qty < 10]
[1] "Book" "Bottle"
> summary(value)
Min. 1st Qu. Median Mean 3rd Qu. Max.
200.0 237.5 275.0 2062.5 2100.0 7500.0

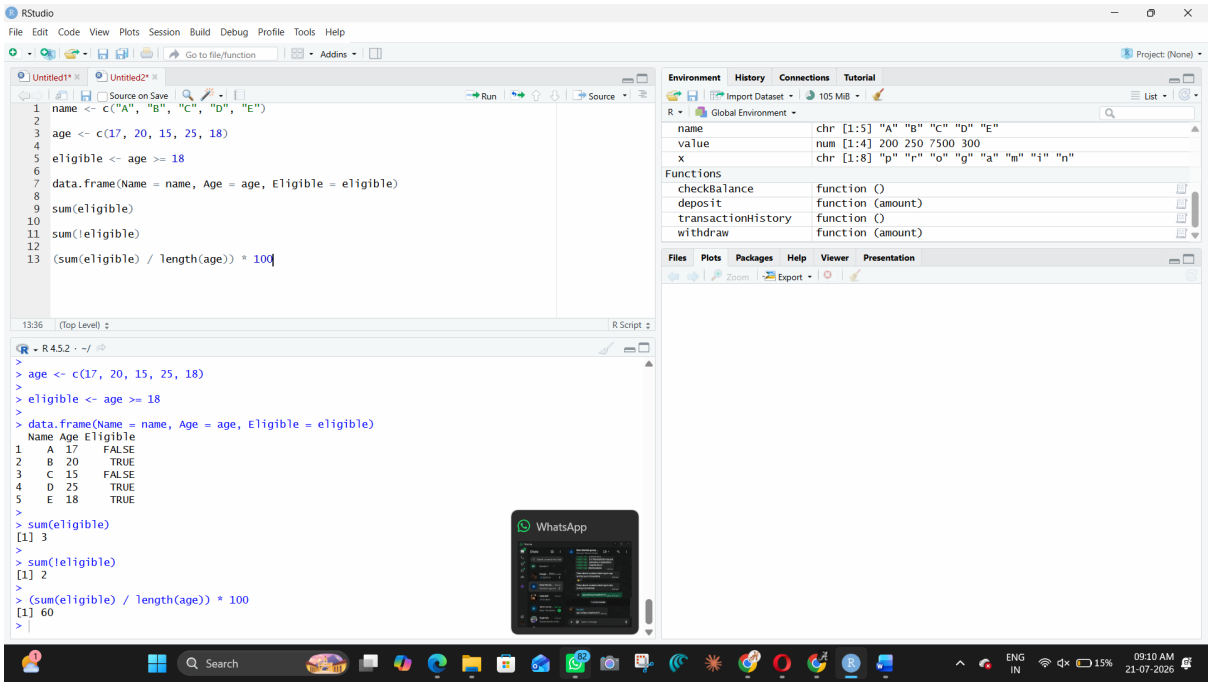
5)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
patient <- data.frame(
Age = c(25, 65, 40, 72),
Gender = c("M", "F", "F", "M"),
Cost = c(20000, 50000, 30000, 45000)
)
mean(patient$Cost)
patient[patient$Age > 60,]
summary(patient)
Environment History Connections Tutorial
patient 4 obs. of 3 variables
product List of 4
temp num [1:7, 1:3] 30 32 31 29 33 30 31 34 32 28 ...
Values
avg 81.9666666666667
balance 1200
history chr [1:2] "Deposited 500" "Withdrawn 300"
value num [1:4] 200 250 7500 300
Files Plots Packages Help Viewer Presentation
R 4.5.2 · ~/
> mean(patient$Cost)
[1] 36250
> patient[patient$Age > 60,]
Age Gender Cost
2 65 F 50000
4 72 M 45000
> summary(patient)
Age Gender Cost
Min. :25.00 Length:4 Min. :20000
1st Qu.:36.25 Class :character 1st Qu.:27500
Median :52.50 Mode :character Median :37500
Mean :50.50 Mean :36250
3rd Qu.:66.75 3rd Qu.:46250
Max. :72.00 Max. :50000

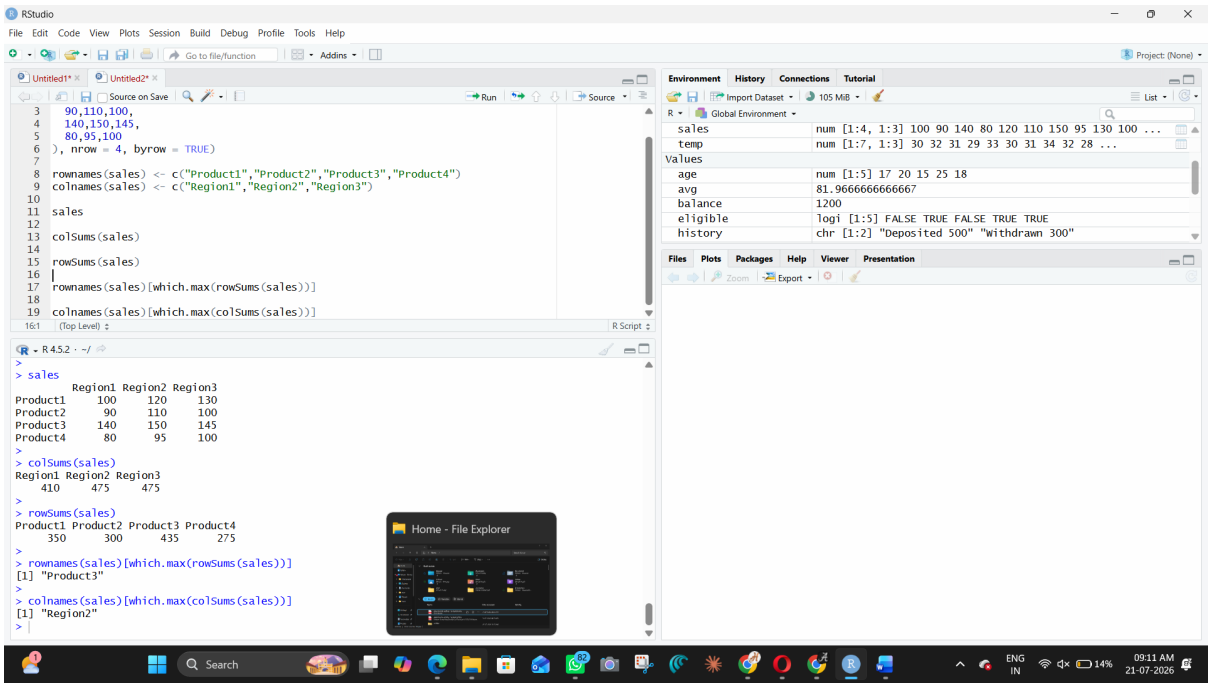
6)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
}
checkBalance <- function() {
print(balance)
}
transactionHistory <- function() {
print(history)
}
deposit(500)
withdraw(300)
checkBalance()
transactionHistory()
Environment History Connections Tutorial
history chr [1:2] "Deposited 500" "Withdrawn 300"
value num [1:4] 200 250 7500 300
x chr [1:8] "p" "r" "o" "g" "a" "m" "i" "n"
Functions
checkBalance function ()
deposit function (amount)
transactionHistory function ()
withdraw function (amount)
Files Plots Packages Help Viewer Presentation
print("Insufficient Balance")
}
}
> checkBalance <- function() {
+ print(balance)
+ }
> transactionHistory <- function() {
+ print(history)
+ }
> deposit(500)
> withdraw(300)
> checkBalance()
[1] 1200
> transactionHistory()
[1] "Deposited 500" "Withdrawn 300"

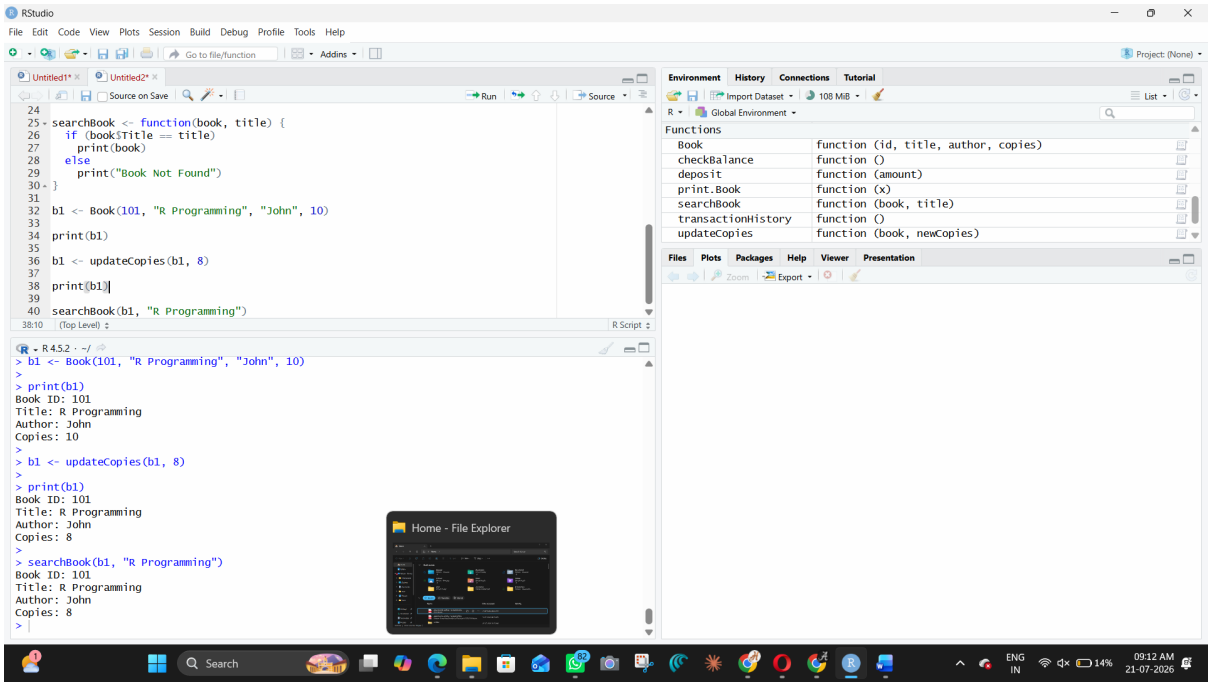
7)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
name <- c("A", "B", "C", "D", "E")
age <- c(17, 20, 15, 25, 18)
eligible <- age >= 18
data.frame(Name = name, Age = age, Eligible = eligible)
sum(eligible)
sum(!eligible)
(sum(eligible) / length(age)) * 100
R 4.5.2 · ~/
> age <- c(17, 20, 15, 25, 18)
> eligible <- age >= 18
> data.frame(Name = name, Age = age, Eligible = eligible)
Name Age Eligible
1 A 17 FALSE
2 B 20 TRUE
3 C 15 FALSE
4 D 25 TRUE
5 E 18 TRUE
> sum(eligible)
[1] 3
> sum(!eligible)
[1] 2
> (sum(eligible) / length(age)) * 100
[1] 60
Environment History Connections Tutorial
name chr [1:5] "A" "B" "C" "D" "E"
value num [1:4] 200 250 7500 300
x chr [1:8] "p" "r" "o" "g" "a" "m" "i" "n"
Functions
checkBalance function ()
deposit function (amount)
transactionHistory function ()
withdraw function (amount)
Files Plots Packages Help Viewer Presentation
WhatsApp
Search
ENG IN
15%
09:10 AM 21-07-2026

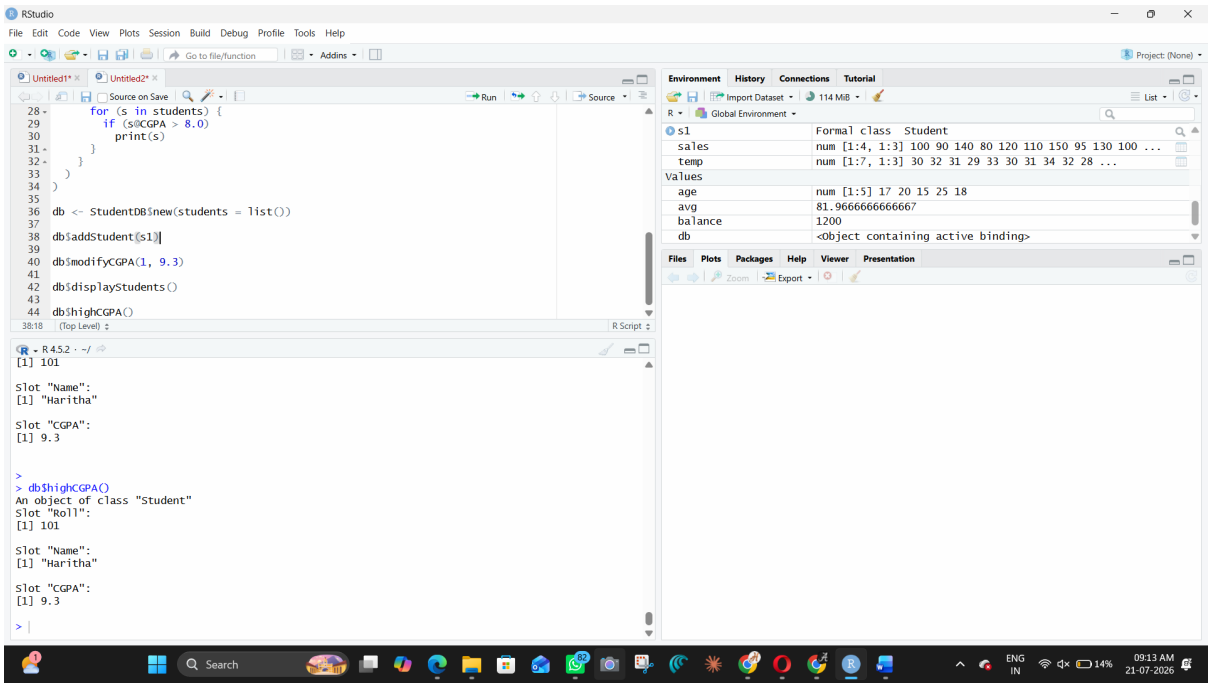
8)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
90,110,100,
140,150,145,
80,95,100
), nrow = 4, byrow = TRUE)
rownames(sales) <- c("Product1","Product2","Product3","Product4")
colnames(sales) <- c("Region1","Region2","Region3")
sales
colSums(sales)
rowSums(sales)
rownames(sales)[which.max(rowSums(sales))]
colnames(sales)[which.max(colSums(sales))]
R 4.5.2 · ~/
> sales
Region1 Region2 Region3
Product1 100 120 130
Product2 90 110 100
Product3 140 150 145
Product4 80 95 100
> colSums(sales)
Region1 Region2 Region3
410 475 475
> rowSums(sales)
Product1 Product2 Product3 Product4
350 300 435 275
> rownames(sales)[which.max(rowSums(sales))]
[1] "Product3"
> colnames(sales)[which.max(colSums(sales))]
[1] "Region2"
Environment History Connections Tutorial
sales num [1:4, 1:3] 100 90 140 80 120 110 150 95 130 100 ...
temp num [1:7, 1:3] 30 32 31 29 33 30 31 34 32 28 ...
Values
age num [1:5] 17 20 15 25 18
avg 81.9666666666667
balance 1200
eligible logi [1:5] FALSE TRUE FALSE TRUE TRUE
history chr [1:2] "Deposited 500" "Withdrawn 300"
Files Plots Packages Help Viewer Presentation
Home - File Explorer
Search
ENG IN
14%
09:11 AM 21-07-2026

9)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
Source on Save Run Source
searchBook <- function(book, title) {
if (book$Title == title)
print(book)
else
print("Book Not Found")
}
b1 <- Book(101, "R Programming", "John", 10)
print(b1)
b1 <- updateCopies(b1, 8)
print(b1)
searchBook(b1, "R Programming")
Environment History Connections Tutorial
Import Dataset 108 MiB
Global Environment
Functions
Book function (id, title, author, copies)
checkBalance function ()
deposit function (amount)
print.Book function (x)
searchBook function (book, title)
transactionHistory function ()
updateCopies function (book, newCopies)
Files Plots Packages Help Viewer Presentation
R 4.5.2 · ~/
> b1 <- Book(101, "R Programming", "John", 10)
> print(b1)
Book ID: 101
Title: R Programming
Author: John
Copies: 10
> b1 <- updateCopies(b1, 8)
> print(b1)
Book ID: 101
Title: R Programming
Author: John
Copies: 8
> searchBook(b1, "R Programming")
Book ID: 101
Title: R Programming
Author: John
Copies: 8
Home - File Explorer
Search
ENG IN
14%
09:12 AM
21-07-2026

10)


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2*
Source on Save Run Source
for (s in students) {
if (s@CGPA > 8.0)
print(s)
}
}
)
)
db <- StudentDB$new(students = list())
db$addStudent(s1)
db$modifyCGPA(1, 9.3)
db$displayStudents()
db$highCGPA()
Environment History Connections Tutorial
Import Dataset 114 MiB
Global Environment
s1 Formal class Student
sales num [1:4, 1:3] 100 90 140 80 120 110 150 95 130 100 ...
temp num [1:7, 1:3] 30 32 31 29 33 30 31 34 32 28 ...
Values
age num [1:5] 17 20 15 25 18
avg 81.9666666666667
balance 1200
db <Object containing active binding>
Files Plots Packages Help Viewer Presentation
R 4.5.2 · ~/
[1] 101
Slot "Name":
[1] "Haritha"
Slot "CGPA":
[1] 9.3
> db$highCGPA()
An object of class "Student"
Slot "Roll":
[1] 101
Slot "Name":
[1] "Haritha"
Slot "CGPA":
[1] 9.3
Search
ENG IN
14%
09:13 AM
21-07-2026